

Task 4 - Backpropagation

Forward Pass

1. Hidden Layer Inputs

$$z_1 = 0.6 \times 0.5 + 0.4 \times 0.5 = 0.5$$

$$z_2 = 0.6 \times 0.5 + 0.4 \times 0.5 = 0.5$$

Hidden Layer Outputs (Sigmoid):

$$a_1 = a_2 = \sigma(0.5) = \frac{1}{1 + e^{-0.5}} \approx 0.6225$$

2. Output Layer Input

$$z_{\text{out}} = 0.6225 \times 0.5 + 0.6225 \times 0.5 = 0.6225$$

Final Output (Sigmoid):

$$\hat{y} = \sigma(0.6225) \approx 0.6502$$

3. Loss (MSE):

$$\mathcal{L} = (\hat{y} - y_{\text{true}})^2 = (0.6502 - 1)^2 \approx 0.1223$$

Backward Pass

1. Output Node Gradient

- $\frac{\partial \mathcal{L}}{\partial \hat{y}} = 2(\hat{y} - y_{\text{true}}) \approx -0.6984$
- $\sigma'(z_{\text{out}}) = \hat{y}(1 - \hat{y}) \approx 0.2273$
- $\delta_{\text{out}} = -0.6996 \times 0.2275 \approx -0.1587$

2. Hidden-to-Output Weight Gradients:

$$\frac{\partial \mathcal{L}}{\partial w_5} = \frac{\partial \mathcal{L}}{\partial w_6} = \delta_{\text{out}} \times a_1 \approx -0.0989$$

3. Hidden Node Gradients:

$$\delta_{h1} = \delta_{h2} = \delta_{\text{out}} \times 0.5 \times \sigma'(0.5) \approx -0.0187$$

4. Input-to-Hidden Weight Gradients:

- $\frac{\partial \mathcal{L}}{\partial w_1} = \delta_{h1} \times 0.6 \approx -0.0112$
- $\frac{\partial \mathcal{L}}{\partial w_2} = \delta_{h1} \times 0.4 \approx 0.0075$
- $\frac{\partial \mathcal{L}}{\partial w_3}, \frac{\partial \mathcal{L}}{\partial w_4}$ are identical to w_1, w_2 .

5. Weight Updates (Learning Rate $\eta = 0.1$):

- w_5, w_6 : $0.5 - 0.1 \times (-0.0989) \approx 0.5099$
- w_1, w_3 : $0.5 - 0.1 \times (-0.0112) \approx 0.5011$
- w_2, w_4 : $0.5 - 0.1 \times (-0.0075) \approx 0.5008$